

Worksheet-1



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Tasks

Task 1 : Programming Exercises:[Data types and Conditional Statements]

1. Write a program that takes a temperature value from the user. It should then allow the user to choose between Celsius (C) and Fahrenheit (F) for conversion. After the user selection, it should then convert the entered temperature to the chosen scale and display the result. Use appropriate data types for temperature and handle error like non-numeric input. Use the following formula for conversion:
$$F = (C \times 9/5) + 32$$
$$C = (F - 32) \times 5/9$$

```
#include<iostream>
using namespace std;

int main()
{
    float ip, op, a;
    cout << "What do you want to perform?" << endl
        << "1. convert into ce" << endl
        << "2. convert into fa" << endl;
    cin >> a;

    if (a == 1)
    {
        cout << "Temperature in fa = ";
        cin >> ip;
        op = (ip - 32) * 5 / 9;
        cout << "Temperature in ce = " << op << endl;
    }
    else
    {
        cout << "Temperature in ce = ";
        cin >> ip;
        op = (ip * 9 / 5) + 32;
        cout << "Temperature in fa = " << op << endl;
    }

    return 0;
}
```

Output:

```
"C:\Users\ashr\Desktop\Co" X + v
What do you want to perform?
1. convert into ce
2. convert into fa
1
Temperature in fa = 132
Temperature in ce = 55.5556

Process returned 0 (0x0)   execution time : 9.803 s
Press any key to continue.
|
```

2. Write a C++ program to implement a number guessing game with different difficulty levels. Easy difficulty ranges from 1-8, medium from 1-30, hard from 1-50. Then, generate a random number to check if the guess is correct based on the user's selection.

```
#include <iostream>
#include <cstdlib> // for rand(), srand()
#include <ctime>   // for time()
using namespace std;

int main()
{
    int ch, guess, randomnumber, maxRange;

    cout << "Choose a difficulty level:" << endl;
    cout << "1. Easy (1-8)" << endl;
    cout << "2. Medium (1-30)" << endl;
    cout << "3. Hard (1-50)" << endl;
    cout << "Enter your choice (1-3): ";
    cin >> ch;

    if (ch == 1)
        maxRange = 8;
    else if (ch == 2)
        maxRange = 30;
    else if (ch == 3)
        maxRange = 50;
    else
    {
        cout << "Invalid choice. Exiting program." << endl;
        return 1;
    }

    srand(time(0));
    randomnumber = (rand() % maxRange) + 1;

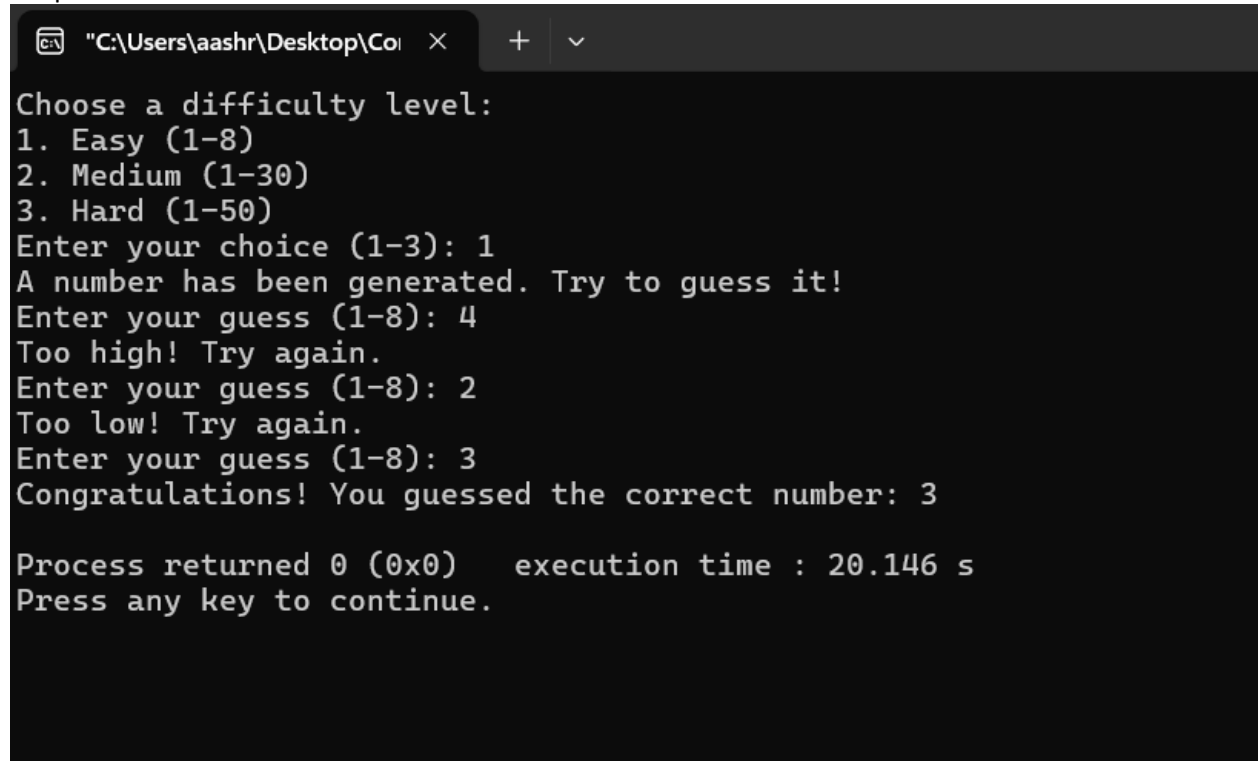
    cout << "A number has been generated. Try to guess it!" << endl;

    do
    {
        cout << "Enter your guess (1-" << maxRange << "): ";
        cin >> guess;

        if (guess > randomnumber)
            cout << "Too high! Try again." << endl;
        else if (guess < randomnumber)
            cout << "Too low! Try again." << endl;
        else
            cout << "Congratulations! You guessed the correct number: " << randomnumber << endl;
    }
```

```
    }  
    while (guess != randomnumber);  
  
    return 0;  
}
```

Output



```
"C:\Users\aaashr\Desktop\Col" X + v
Choose a difficulty level:
1. Easy (1-8)
2. Medium (1-30)
3. Hard (1-50)
Enter your choice (1-3): 1
A number has been generated. Try to guess it!
Enter your guess (1-8): 4
Too high! Try again.
Enter your guess (1-8): 2
Too low! Try again.
Enter your guess (1-8): 3
Congratulations! You guessed the correct number: 3

Process returned 0 (0x0)   execution time : 20.146 s
Press any key to continue.
```

3. Write a program that reads an array of integer numbers from the user and sorts the numbers in the ascending order.

```
#include<iostream>
using namespace std;
int main()
{
    int i;
    int a[100];
    int temp=0;
    cout<<"Enter 10 numbers"<<endl;
    for(int i=0;i<=9;i++)
    {
        cin>>a[i];
```

```
    }  
    for (int i=0;i<=9;i++)  
    {  
        for (int j=i+1;j<8;j++)  
  
            if(a[i]< a[j])  
            {  
                temp=a[i];  
                a[i]=a[j];  
                a[j]=temp;  
            }  
        }  
        cout<<"The numbers in ascending order is: "<<endl;  
  
        for (int i=0;i<10;i++)  
        {  
            cout<<a[i]<<endl;  
  
        }  
        return 0;  
    }
```

Output:

```
"C:\Users\aaashr\Desktop\Co... X + v
Enter 10 numbers
1
4
6
8
3
1
2
5
8
0
The numbers in ascending order is:
8
6
5
4
3
2
1
1
8
0
Process returned 0 (0x0)   execution time : 10.082 s
Press any key to continue.
|
```


4. Write a program that reads a number from the user and based on the user input, it says what day of the week it is, Sundays being 1 and Saturdays being 7. Your system should give appropriate response for invalid input entries.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int day;
```

```
    cout << "Enter a number:" << endl;
```

```
    cout << "1 = Sunday" << endl;
```

```
    cout << "2 = Monday" << endl;
```

```
    cout << "3 = Tuesday" << endl;
```

```
    cout << "4 = Wednesday" << endl;
```

```
    cout << "5 = Thursday" << endl;
```

```
    cout << "6 = Friday" << endl;
```

```
    cout << "7 = Saturday" << endl;
```

```
    cin >> day;
```

```
    switch(day) {
```

```
        case 1:
```

```
            cout << "Sunday" << endl;
```

```
            break;
```

```
        case 2:
```

```
            cout << "Monday" << endl;
```

```
            break;
```

```
        case 3:
```

```
            cout << "Tuesday" << endl;
```

```
            break;
```

```
        case 4:
```

```
            cout << "Wednesday" << endl;
```

```
            break;
```

```
        case 5:
```

```
            cout << "Thursday" << endl;
```

```
            break;
```

```
        case 6:
```

```
            cout << "Friday" << endl;
```

```
            break;
```

```
        case 7:
```

```
            cout << "Saturday" << endl;
```

```
            break;
```

```
        default:
```

```
            cout << "Sorry!,That's not a valid choice. Try a number from 1 to 7." << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

Output:

```
"C:\Users\Aashr\Desktop\Co...  X  +  v
Enter a number:
1 = Sunday
2 = Monday
3 = Tuesday
4 = Wednesday
5 = Thursday
6 = Friday
7 = Saturday
7
Saturday

Process returned 0 (0x0)   execution time : 10.329 s
Press any key to continue.
|
```

Task 2: Programming Exercises: [Control Statements]

1. Create a program that takes a positive integer as input and determines whether it's a "bouncy number". A bouncy number is one where the digits neither consistently increase nor consistently decrease when read from left to right.

```
#include <iostream>

using namespace std;

int main()
{
    int number;
    int D1;
    int D2;
    cout<<"Enter a number"<<endl;
    cin>>number;

    if(number<100)
    {
        cout<<"out of range! Bouncy numbers has atleast 3 digit"<<endl;
    }
    bool increasing=false;
    bool decreasing=false;

    D1= number % 10;
    number = number / 10;

    while (number > 0)
    {
        D2 = number % 10;

        if (D2 < D1)
        {
            increasing = true;
        }
        else if (D2 > D1)
        {
            decreasing = true;
        }

        D1 = D2;
        number = number / 10;
    }
}
```

```
}

if (increasing && decreasing)
{
    cout << "It is a BOUNCY number." << endl;

}
else
{
    cout << "It is not a Bouncy number." << endl;
}

return 0;
}
```

Output:

```
"C:\Users\ashr\Desktop\Co" × + v
Enter a number
132
It is a BOUNCY number.

Process returned 0 (0x0)   execution time : 2.813 s
Press any key to continue.
|
```

Task 3: Programming Exercises on Arrays

1. Write a program that manages a cinema ticket booking system. The program should display a 5x5 seating arrangement where:

1. Available seats are marked with 'O'
2. Booked seats are marked with 'X'

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    char matA[5][5];
```

```
    cout << "Seating Arrangement:\n";
```

```
    for (int i = 0; i < 5; i++)
```

```
    {
```

```
        for (int j = 0; j < 5; j++)
```

```
        {
```

```
            matA[i][j] = 'O';
```

```
            cout << " " << matA[i][j] << " ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    char choice;
```

```
    do
```

```
    {
```

```
        int row, col;
```

```
        cout << "Enter the row (0-4) to book: ";
```

```
        cin >> row;
```

```
cout << "Enter the column (0-4) to book: ";
cin >> col;

if (row >= 0 && row < 5 && col >= 0 && col < 5)
{
    if (matA[row][col] == 'O')
    {
        matA[row][col] = 'X';
        cout << "Seat booked successfully!\n";
    }
    else
    {
        cout << "Sorry, this seat is already booked!\n";
    }
}
else
{
    cout << "Invalid seat position!\n";
}

cout << "Updated Seating Arrangement:\n";
for (int i = 0; i < 5; i++)
{
    for (int j = 0; j < 5; j++)
    {
        cout << " " << matA[i][j] << " ";
    }
    cout << endl;
}
```

```
cout << "Do you want to book another seat? (y/n): ";
cin >> choice;

}

while (choice == 'y' || choice == 'Y');

cout << "Final Seating Arrangement:\n";
for (int i = 0; i < 5; i++)
{
    for (int j = 0; j < 5; j++)
    {
        cout << " " << matA[i][j] << " ";
    }
    cout << endl;
}

cout << "Thank you for booking!\n";

return 0;
}
```


Output:

```
"C:\Users\aaashr\Desktop\Co... X + v
Seating Arrangement:
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
Enter the row (0-4) to book: 4
Enter the column (0-4) to book: 4
Seat booked successfully!
Updated Seating Arrangement:
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 X
Do you want to book another seat? (y/n): y
Enter the row (0-4) to book: 1
Enter the column (0-4) to book: 1
Seat booked successfully!
Updated Seating Arrangement:
0 0 0 0 0
0 X 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 X
Do you want to book another seat? (y/n): |
```